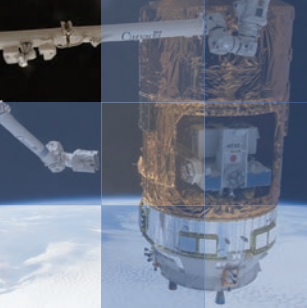




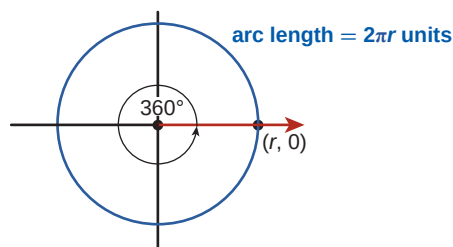
Radians and Degrees

Radian measure is related to the length of the arc of a circle that is intercepted by an angle.

Determining Arc Length

Arc length is part of the circumference of a circle. The arc length of a full circle, which measures 360° , is the same as the circle's circumference of $2\pi r$ units.

For an arc intercepted by an angle of any measure, you can use proportionality to determine its length.

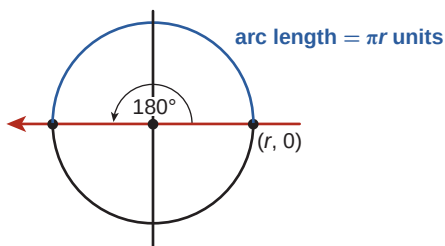


Arc Length Proportionality Property

The length of an arc is proportional to the measure of its central angle.

An arc intercepted by a 180° angle is a semicircle, so its length is equal to one-half the circumference of a circle. That length is $s = \frac{1}{2}(2\pi r) = \pi r$. You can also use a proportion to determine the length between the arc length s and the measure of the central angle, 180° .

$$\begin{aligned}\frac{\text{arc length}}{2\pi r} &= \frac{\text{degrees}}{360^\circ} \\ \frac{s}{2\pi r} &= \frac{180^\circ}{360^\circ} \\ 2\pi r \cdot \frac{s}{2\pi r} &= \frac{180^\circ}{360^\circ} \cdot 2\pi r \\ s &= \frac{1}{2} \cdot 2\pi r \\ s &= \pi r\end{aligned}$$



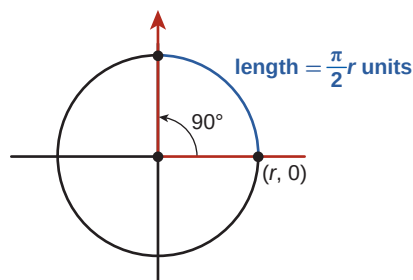
EXAMPLE 1

What is the length s of the arc on a circle with a radius of r units that is intercepted by a central angle of 90° ?

SOLUTION

An arc intercepted by a 90° angle is a quarter circle, so its length is equal to one-fourth the circumference of the circle.

That length is $s = \frac{1}{4}(2\pi r) = \frac{\pi}{2}r$. ■



► **Remember** You can also use a proportion to determine the arc length of the 90° angle.

$$\begin{aligned}\frac{s}{2\pi r} &= \frac{90^\circ}{360^\circ} \\ 2\pi r \cdot \frac{s}{2\pi r} &= \frac{90^\circ}{360^\circ} \cdot 2\pi r \\ s &= \frac{1}{4} \cdot 2\pi r \\ s &= \frac{\pi}{2}r\end{aligned}$$

EXAMPLE 2

What is the measure of the central angle on a circle with radius 40 cm that intercepts a 120 cm arc?

► **Remember** Arc length is a distance; it is part of the circumference of a circle.

SOLUTION

Use a proportion between the arc length and the unknown angle θ .

$$\begin{aligned}\frac{\theta}{360^\circ} &= \frac{120}{2\pi(40)} && \text{Set up the proportion.} \\ 360^\circ \cdot \frac{\theta}{360^\circ} &= \frac{120}{80\pi} \cdot 360^\circ && \text{Multiply both sides by } 360^\circ. \\ \theta &\approx 171.89^\circ && \text{Use a calculator to simplify.}\end{aligned}$$

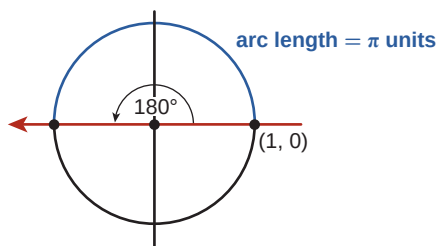
The angle measure is approximately 171.89° . ■

Defining Radian Measure

Definition

The **unit circle** is a circle with a radius of one unit.

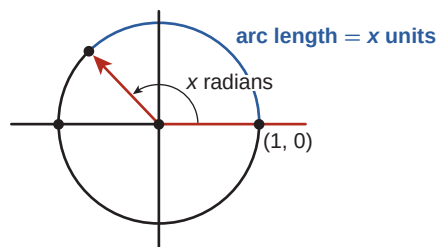
The circumference of a unit circle is $2\pi(1)$, or 2π units. It is the same as the arc length on a unit circle with a central angle of 360° . Now look at an arc that is a semicircle. It is intercepted by a 180° angle. The length of the arc is one-half of the circumference of the circle. One-half of 2π is π . So an arc on the unit circle with a 180° central angle has a length of π units.



Definition

The **radian** measure of an angle is the length of the arc on a unit circle subtended (intercepted) by the angle.

If an arc on the unit circle intercepted by an angle has a length of x units, then the angle itself is said to have a measure of x radians. Therefore, because the arc length of one full unit circle is 2π , the radian measure of 360° is 2π .



EXAMPLE 3

What is the radian measure x of an angle that intercepts a quarter circle of radius 1?

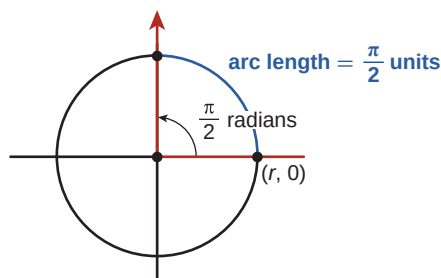
► Q&A

Q What is the degree measure of a quarter circle?

A 90°

SOLUTION

The angle has a measure in radians that is equal to $\frac{1}{4}$ of the radian measure of the entire unit circle. The radian measure of the full circle is 2π . Take $\frac{1}{4}$ of 2π to get $x = \frac{1}{4}(2\pi) = \frac{\pi}{2}$ radians. ■



Converting Between Degrees and Radians

If you know an angle's degree measure, you can figure out its radian measure and vice versa.

The radian measure of an entire circle, which has a degree measure of 360° , is equal to 2π . The radian measure of one-half of a circle, which measures 180° , is equal to one-half of 2π , which is π . Just as degree measure and arc length are proportional, degree measure and radian measure are proportional.

Degrees-Radians Proportionality Property

Let θ be the measure of an angle in degrees and x be the measure of the angle in radians.

$$\frac{\theta}{180^\circ} = \frac{x}{\pi}$$

You can solve the proportion for degrees (θ) by multiplying both sides by 180° to get $\theta = \frac{x}{\pi}(180^\circ) = x\left(\frac{180^\circ}{\pi}\right)$. You can likewise solve for radians (x) by multiplying both sides by π to get $x = \frac{\theta}{180^\circ}(\pi) = \theta\left(\frac{\pi}{180^\circ}\right)$.

How to Convert Between Degrees and Radians

To convert radians to degrees, multiply a radian measure by $\frac{180^\circ}{\pi}$.

To convert degrees to radians, multiply a degree measure by $\frac{\pi}{180^\circ}$.

EXAMPLE 4

- A** What is the measure of -135° in radians? Give your answer in terms of π .

SOLUTION

Multiply by $\frac{\pi}{180^\circ}$ and simplify.

$$-135^\circ \left(\frac{\pi}{180^\circ} \right) = -\frac{3\pi}{4}$$

- **Think About It** A counterclockwise rotation generates a positive angle measure (in degrees or radians), while a clockwise rotation generates a negative measure.

- B** What is the measure of $\frac{5\pi}{6}$ in degrees?

SOLUTION

Multiply by $\frac{180^\circ}{\pi}$ and simplify.

$$\frac{5\pi}{6} \left(\frac{180^\circ}{\pi} \right) = \frac{5}{1} \left(\frac{30^\circ}{1} \right) = 150^\circ \quad \blacksquare$$

- **Think About It** When no unit of measure is specified for an angle, radian measure is understood. For example, $\frac{\pi}{6}$ and 3.5 are both radian measures.

EXAMPLE 5

Determine the indicated measure.

- **Remember** Complementary angles have measures that sum to 90° . Supplementary angles have measures that sum to 180° .

A the complement of $\frac{3\pi}{8}$

SOLUTION

First determine the measure of a 90° angle in radians.

$$90^\circ \left(\frac{\pi}{180^\circ} \right) = \frac{\pi}{2}$$

Then subtract $\frac{3\pi}{8}$ from that measure.

$$\frac{\pi}{2} - \frac{3\pi}{8} = \frac{4\pi}{8} - \frac{3\pi}{8} = \frac{\pi}{8}$$

B the supplement of $\frac{3\pi}{8}$

SOLUTION

First determine the measure of a 180° angle in radians.

$$180^\circ \left(\frac{\pi}{180^\circ} \right) = \pi$$

Then subtract $\frac{3\pi}{8}$ from that measure.

$$\pi - \frac{3\pi}{8} = \frac{8\pi}{8} - \frac{3\pi}{8} = \frac{5\pi}{8} \quad \blacksquare$$

► **Think About It** Unless otherwise indicated, always give measures in the units provided in a problem.